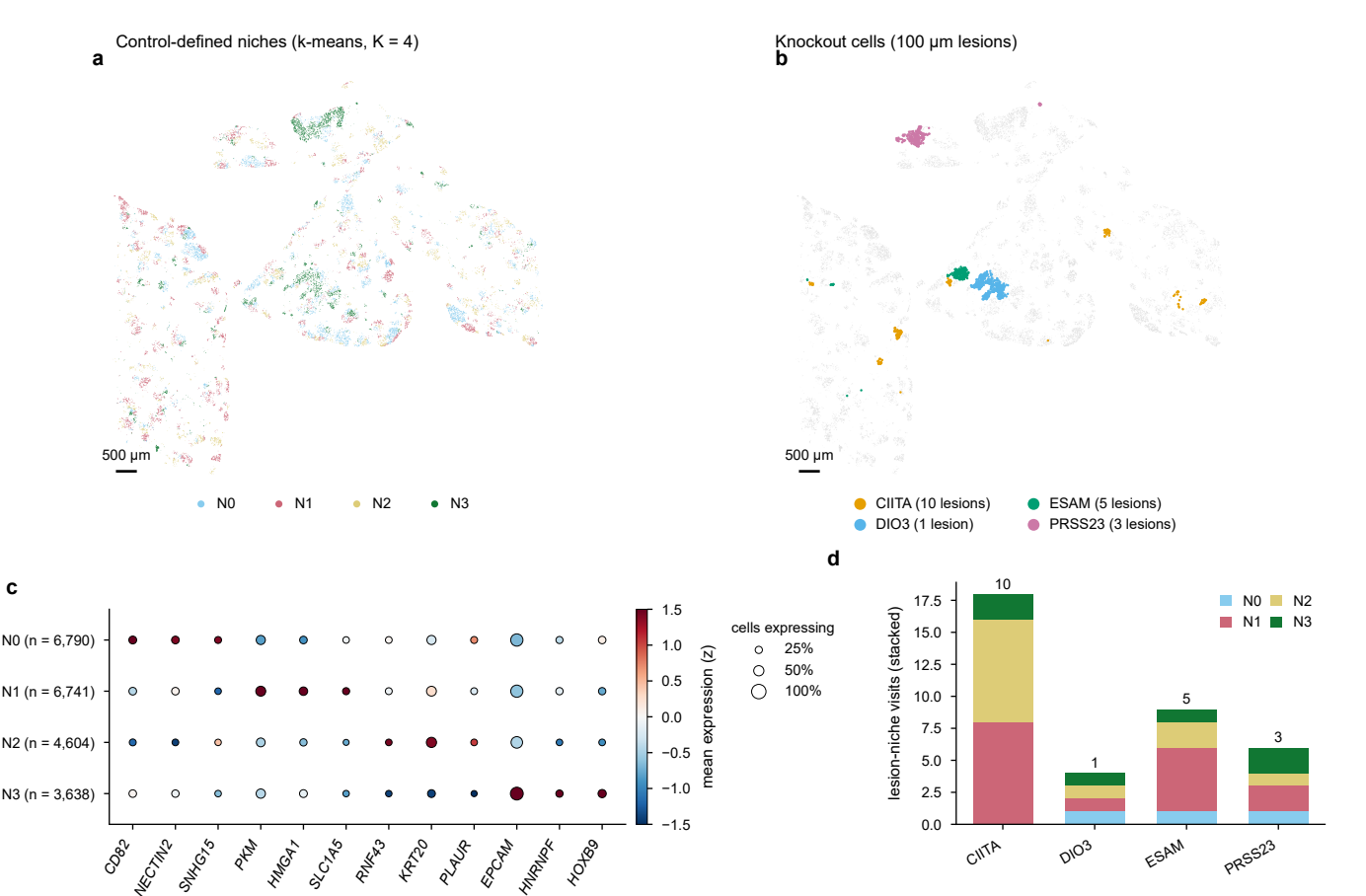

a
Control-defined niches (k-means, K = 4)
500 µm
N0
N1
N2
N3
b
Knockout cells (100 µm lesions)
500 µm
CIITA (10 lesions)
DIO3 (1 lesion)
ESAM (5 lesions)
PRSS23 (3 lesions)
c
N0 (n = 6,790)
N1 (n = 6,741)
N2 (n = 4,604)
N3 (n = 3,638)
CD82
NECTIN2
SNHG15
PKM
HMGA1
SLC1A5
RNF43
KRT20
PLAUR
EPCAM
HNRNPF
HOXB9
mean expression (z)
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
cells expressing
25%
50%
100%
d
lesion-niche visits (stacked)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
10
1
5
3
N0
N1
N2
N3
CIITA
DIO3
ESAM
PRSS23